

Log Analysis: Java MDC Context & Distributed Transaction Tracing

Duration : 60 Min.

Scenario

A distributed order checkout flow failed across 4 microservices. Trace the MDC span context to reconstruct the exact failure timeline.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Fill out 'trace_timeline.json' with chronological service span sequence and failing service name.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running `python run.py` locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	trace_timeline.json exists and is valid JSON	5 Marks
TC2	Trace ID ('trace-7741') correctly matched across all spans	5 Marks
TC3	Failing downstream span service correctly identified	5 Marks

TC4	Span execution latency correctly calculated	5 Marks
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Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.